Lecture 23.0

Monday, 9 November 2026 22:03

Real intervals.

Definition: Let $a, b \in R^*$. Then

① Closed intervals $[a,b] := \{x \in R^* : a \le x \le b\}$

② Half open intervals $(a,b] := \{x \in R^* : a < x \le b\}$

$[a,b) := \{x \in R^* : a \le x < b\}$

③ Open intervals $(a,b) := \{x \in R^* : a < x < b\}$.

Adherent points of a set: Let $X \subseteq R$ and $x \in R$. We say that x is an adherent point of X iff

$\forall \varepsilon > 0 \quad \exists y \in X$ s.t. $|x-y| \le \varepsilon$.

Remark: Each $x \in X$ is an adherent point of X as take $y = x \in X$ s.t. $|x-y| = 0 < \varepsilon$ for all $\varepsilon > 0$.

Claim: a and b both are adherent points of (a,b).

Proof: Fix $\varepsilon > 0$. Let $\delta = \min\{\varepsilon, \frac{b-a}{2}\} > 0$

For $a \in R$, choose $y = a + \delta > a$. Further, $y = a + \delta < a + \frac{b-a}{2} = \frac{a+b}{2} < b$

$\Rightarrow y \in (a,b)$ and $|y-a| = \delta \le \varepsilon$.

For $b \in R$, choose $y = b - \delta$. Then.

$y \in (a,b)$ and $|y-b| = \delta \le \varepsilon$.

Remark: Let $c < a$, then c is not an adherent point of (a,b).

Take $\varepsilon = a - c > 0$. For any $y \in (a,b)$, $y > a$ so

$|y-c| = y - c > a - c = \varepsilon$

$\Rightarrow$ c is not an adherent point of (a,b), when $c < a$.

Similarly, $d > b$ is also not an adherent point of (a,b).

Closure of a set: Let $X \subseteq R$. Then closure of X, denoted as $\overline{X}$, is defined as the collection of all adherent points of X. That is

$\overline{X} := \{x : x \text{ is an adherent point of } X\}$.

Example: $\overline{(a,b)} = (a,b) \cup \{a,b\} = [a,b]$.

Lemma: Let $X, Y \subseteq R$. Then

① $X \subseteq \overline{X}$

② If $X \subseteq Y \Rightarrow \overline{X} \subseteq \overline{Y}$.

③ $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$

④ $\overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$

Proof: ① Let $x \in X$ and $\varepsilon > 0$. Choose $y = x \in X$ s.t. $|y-x| = 0 \le \varepsilon$.

$\Rightarrow x \in \overline{X}. \Rightarrow X \subseteq \overline{X}$.

② $X \subseteq Y \Rightarrow \overline{X} \subseteq \overline{Y}$.

Let $x \in \overline{X}$ and $\varepsilon > 0$. Then

$\exists y \in X$ s.t. $|x-y| \le \varepsilon$.

Since $X \subseteq Y \Rightarrow y \in Y$

$\Rightarrow$ x is an adherent point of Y.

$\Rightarrow x \in \overline{Y} \quad \Rightarrow \overline{X} \subseteq \overline{Y}$.

③ $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$

($\supseteq$) Since $X \subseteq X \cup Y \Rightarrow \overline{X} \subseteq \overline{X \cup Y}$

Since $Y \subseteq X \cup Y \Rightarrow \overline{Y} \subseteq \overline{X \cup Y}$

$\Rightarrow \overline{X} \cup \overline{Y} \subseteq \overline{X \cup Y}$

($\subseteq$) Let $x \in \overline{X \cup Y}$. Suppose for contradiction $x \notin \overline{X}$ and $x \notin \overline{Y}$.

① Since $x \notin \overline{X} \Rightarrow \exists \varepsilon_1 > 0$ s.t. $|x-y| > \varepsilon_1 \;\forall y \in X$.

② Since $x \notin \overline{Y} \Rightarrow \exists \varepsilon_2 > 0$ s.t. $|x-y| > \varepsilon_2 \;\forall y \in Y$

Let $\varepsilon = \min\{\varepsilon_1, \varepsilon_2\} > 0$

Since $x \in \overline{X \cup Y} \Rightarrow \exists z \in X \cup Y$ s.t. $|x-z| \le \varepsilon$.

If $z \in X$, then $|z-x| \le \varepsilon \le \varepsilon_1$ [Contradiction to ①]

If $z \in Y$ then $|z-x| \le \varepsilon \le \varepsilon_2$ [" " ②]

$\Rightarrow x \in \overline{X}$ or $x \in \overline{Y} \Rightarrow x \in \overline{X} \cup \overline{Y}$

i.e. $\overline{X \cup Y} \subseteq \overline{X} \cup \overline{Y}$

Combining both directions $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$.

④ $\overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$.

$X \cap Y \subseteq X \Rightarrow \overline{X \cap Y} \subseteq \overline{X}$

$X \cap Y \subseteq Y \Rightarrow \overline{X \cap Y} \subseteq \overline{Y}$

$\Rightarrow \overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$

[Counter-example to equality]

$X = (0,1)$, $Y = (1,2)$

$X \cap Y = \phi \Rightarrow \overline{X \cap Y} = \phi$

$\overline{X} = [0,1]$, $\overline{Y} = [1,2]$

$\overline{X} \cap \overline{Y} = \{1\} \ne \overline{X \cap Y}$.

Definition (Closed sets): Let $E \subseteq R$. E is said to be closed if $\overline{E} = E$.

or E contains all its adherent points.

Lemma (adherent points as limits):

Let $X \subseteq R$ and $x \in R$. Then x is an adherent point of X iff there exists a sequence $(a_n)_{n \ge 0}$, where $a_n \in X \;\forall n \ge 0$, which converges to x.

Proof: ($\Leftarrow$) Sequence gives adherent point.

Let $(a_n)_{n \ge 0}$ be a sequence such that $a_n \in X \;\forall n \ge 0$

and $\lim_{n \to \infty} a_n = x$.

Let $\varepsilon > 0$. Then $\exists N \ge 0$ s.t. $|a_n - x| \le \varepsilon \;\forall n \ge N$.

Since $y = a_n \in X \Rightarrow |y - x| \le \varepsilon$

$\Rightarrow$ x is an adherent point of X.

($\Rightarrow$ Adherent point gives sequence). Let x be an adherent point of X.

That is $\forall \varepsilon > 0 \quad \exists y \in X$ s.t. $|y-x| \le \varepsilon$.

For each $n \ge 0$ choose $\varepsilon = \frac{1}{n+1}$

Then there is an element $y = a_n \in X$ s.t. $|a_n - x| \le \frac{1}{n+1}$

Since $-\frac{1}{n+1} \le a_n - x \le \frac{1}{n+1} \quad \forall n \ge 0$

From squeeze test (If $a_n \le b_n \le c_n \;\forall n \ge 0$ and $\lim_{n\to\infty} a_n = L = \lim_{n\to\infty} c_n$

$\Rightarrow \lim_{n\to\infty} b_n = 0$)

Above $\Rightarrow \lim_{n\to\infty}(a_n - x) = 0$ or $\lim_{n\to\infty} a_n = x$.

Lemma: Let $X \subseteq R$.

① If $\overline{X} = X$ and $(a_n)_{n \ge 0}$ is a convergent sequence, where $a_n \in X$. Then $\lim_{n\to\infty} a_n \in X$.

② If every convergent sequence $(a_n)_{n \ge 0}$ of elements of X has limit in X, then $\overline{X} = X$.

Proof. ① $\overline{X} = X$. Since $\lim_{n\to\infty} a_n$ exists and let us say it is equal to L. Then L is an adherent point. $\Rightarrow L \in \overline{X} = X$.

② Since $X \subseteq \overline{X}$, then it is sufficient to prove that $\overline{X} \subseteq X$.

Let $L \in \overline{X} \Rightarrow L := \lim_{n\to\infty} a_n$, where $(a_n)_{n \ge 0}$ is some convergent sequence elements of X.

But given $L \in X \Rightarrow \overline{X} \subseteq X$.

§ Lecture 23.1

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Definition (Limit points). Let $X \subseteq R$. We say that

① x is a limit point of X iff it is an adherent point of $X \setminus \{x\}$.

② x is an isolated point of X if $x \in X$ and $\exists\, \varepsilon > 0$ s.t. $|x-y| > \varepsilon$ for all $y \in X \setminus \{x\}$.

Ex. $X = (1,2) \cup \{3\}$.

Clearly each point $x \in [1,2]$ is a limit point of X.

Let $1 < x < 2$, then

$$X = (1,x) \cup [x,2) \cup \{3\}$$

$$X \setminus \{x\} = (1,x) \cup (x,2) \cup \{3\}$$

$\Rightarrow$ x is an adherent point of $(1,x)$ and hence $X \setminus \{x\}$.

Also 3 is an adherent point of X. But it is not a limit point as

$X \setminus \{3\} = (1,2)$

$\exists\, \varepsilon = 0.5$ s.t. $\forall\, y \in X \setminus \{3\}$ $|x-y| = 3-y > 1 > \varepsilon$.

It is an isolated point.

Definition (Bounded set). Let $X \subseteq R$. X is said to be bounded if $X \subseteq [-M, M]$ for some real $M > 0$.

Theorem (Heine-Borel theorem): Let $X \subseteq R$. Then following two statements are equivalent.

① X is closed bounded.

② Given any sequence $(a_n)_{n \geq 0}$ of reals which takes values in X, then exists a subsequence $(a_{n_j})_{j \geq 0}$ of $(a_n)_{n \geq 0}$ that converges to some number $L \in X$.

Proof. ① $\Rightarrow$ ②.

Assume X is closed and bounded. Let $(a_n)_{n \geq 0}$ be some sequence in X.

Since X is bounded and $a_n \in X$ $\forall\, n \geq 0$

$\Rightarrow$ there exists some real number $M > 0$ s.t. $|a_n| \leq M$ $\forall\, n \geq 0$.

$\Rightarrow$ $(a_n)_{n \geq 0}$ is a bounded sequence.

By Bolzano-Weierstrass theorem, there will be a convergent subsequence $(a_{n_j})_{j \geq 0}$ of $(a_n)_{n \geq 0}$. Let $\lim\limits_{j \to \infty} a_{n_j} = L$.

But subsequence also lies in X $\Rightarrow$ $L \in X$ as $X = \overline{X}$.

② $\Rightarrow$ ①

Assume ②.

Proof that X is closed.

Let $(a_n)_{n \geq 0}$ be a convergent sequence of elements of X s.t.

$$\lim_{n \to \infty} a_n = x$$

Since all subsequences of a convergent sequence converge to same number. x.

But from ② there is a subsequence that converge to $L \in X$.

Since limits are unique $x = L \in X$.

Then X is a closed set as all convergent sequences of X converge in X.

Proof that X is bounded.

Suppose X is not bounded.

Then for every $n \geq 0$ $\exists$ an element of X, call it a_n s.t. $|a_n| \geq n$.

From ② some subsequence $(a_{n_j})_{j \geq 0}$ converges to $L \in X$. But a convergent sequence is bounded, i.e. $\exists\, K > 0$ s.t.

$$|a_{n_j}| \leq K \quad \forall\, j \geq 0. \quad \rightarrow ①$$

Then $|a_{n_j}| \geq n_j \geq j \quad \forall j$

Take any $j > K$ then $|a_{n_j}| \geq j > K$ (Contradicting 1)

$\Rightarrow$ X is bounded.

§ Lecture 23.2

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Functions on real line and their restriction:

Let $f : \mathbb{R} \to \mathbb{R}$ be a function.

Let $X \subseteq \mathbb{R}$, then

$f|_X : X \to \mathbb{R}$ is called a restriction of f on X.

$$f|_X(x) = \begin{cases} f(x) & \text{if } x \in X \\ \text{undefined} & \text{if } x \notin X. \end{cases}$$

Convergence of functions at a point.

Let $X \subseteq \mathbb{R}$ and $f : X \to \mathbb{R}$ be a function. Let $E \subseteq X$ and let x_0 be an adherent point of E, and let $L \in \mathbb{R}$. We say that f converges to L at x_0 in E and write

$$\lim_{x \to x_0;\, x \in E} f(x) = L \quad \text{iff}$$

$\forall\, \varepsilon > 0 \quad \exists\, \delta > 0$ s.t. $|f(x) - L| \leq \varepsilon$ for all $x \in X$ such that $|x - x_0| < \delta$.

If f doesnot converge to any number L at x_0, we say that f diverges at x_0.

Theorem: Let $X \subseteq \mathbb{R}$ and $f : X \to \mathbb{R}$ be a function. Let $E \subseteq X$, x_0 be an adherent point of E and $L \in \mathbb{R}$. Then following two statements are equivalent.

① f converges to L at x_0 in E.

② For every sequence $(a_n)_{n \geq 0}$, where $a_n \in E$ and $\lim_{n \to \infty} a_n = x_0$,

$$\lim_{n \to \infty} f(a_n) = L.$$

Proof: ① $\Rightarrow$ ②.

Let $(a_n)_{n \geq 0}$ be a sequence in E s.t. $\lim_{n \to \infty} a_n = x_0$.

Let $\varepsilon > 0$. Since $\lim_{x \to x_0;\, x \in E} f(x) = L$

$\Rightarrow \exists\, \delta > 0$ s.t. $|f(x) - L| \leq \varepsilon$ whenever $x \in E$ and $|x - x_0| < \delta$. $\to$ ①

Since $\lim_{n \to \infty} a_n = x_0 \Rightarrow \exists\, N \geq 0$ s.t. $|a_n - x_0| \leq \delta/2 \;\; \forall\, n \geq N$. $\to$ ②

Now since $a_n \in E$ and $|a_n - x_0| \leq \delta/2 < \delta$, from ①

$$|f(x) - L| \leq \varepsilon$$

$$\Rightarrow \lim_{x \to x_0;\, x \in E} f(x) = L.$$

② $\Rightarrow$ ①.

Assume ② and suppose that $\lim_{x \to x_0;\, x \in E} f(x) \neq L$.

$\Rightarrow \exists\, \varepsilon > 0 \quad \forall\, \delta > 0 \quad \exists\, x \in E$ with $|x - x_0| < \delta$

and $|f(x) - L| > \varepsilon$. $\longrightarrow$ ③

Fix this ε.

Build a bad sequence that converges to x_0.

For each $n \geq 0$, apply ③ with $\delta = \frac{1}{n+1}$

Choose $a_n \in E$ with $|a_n - x_0| < \frac{1}{n+1}$ and $|f(a_n) - L| > \varepsilon$.

The sequence converges to x_0 while f doesn't converge to L.

[Contradiction]